

Impact-Aware State Estimation for Off-Road Motion

GPS / IMU / Camera Fusion with Adaptive NHC Relaxation
via Jerk Heuristic and LSTM

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KEY RESULT

-96%

ATE reduction

The Off-Road State Estimation Challenge



Impulsive Dynamics

Rocks, wheel bounce, steep hills generate 500–3000 m/s^3 jerk — violating smooth-noise assumptions of standard EKF.



NHC-GPS Conflict

Hard NHC zeros vertical velocity every 25 ms. GPS corrections during hill traversal are immediately destroyed. Filter oscillates ± 15 m.

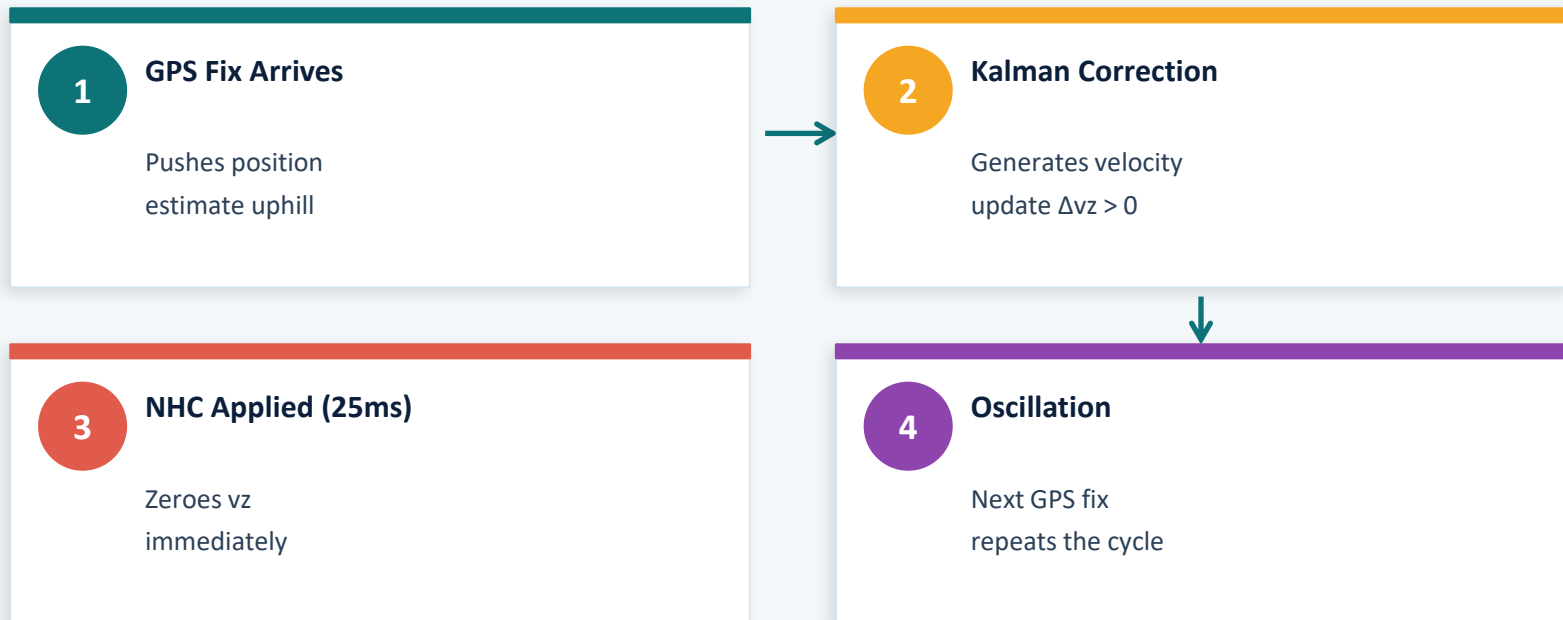


Standard VIO Fails

Baseline GPS/IMU achieves 6.3 m ATE on ROOAD updown sequences — unusable for off-road autonomy.

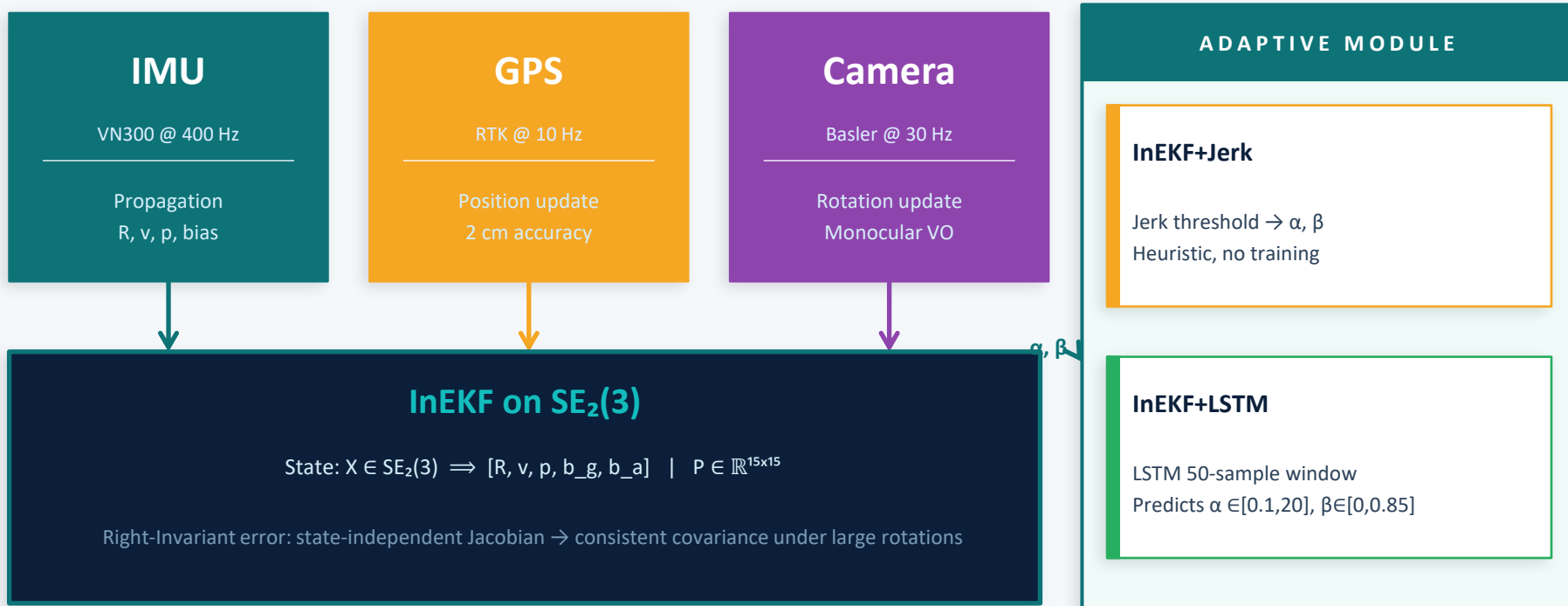


Why the Hard NHC Breaks GPS Fusion on Hills



Solution: Adaptive $\beta > 0$ allows velocity correction to persist → GPS and NHC work together, not against each other

InEKF on $SE_2(3)$ — Three-Sensor Fusion Pipeline



Two Adaptive Mechanisms — Jerk Heuristic vs LSTM

InEKF + Jerk

Heuristic — no training required

Compute smoothed 20-sample jerk magnitude $j(t)$

When $j > 8 \text{ m/s}^3$ (active throughout ROOAD):

$\alpha \leftarrow 15$ (inflate process noise)

$\beta \leftarrow 0.85$ (relax NHC)

Exponential decay when terrain quiets:

$\alpha \leftarrow \max(1, 0.95 \cdot \alpha)$ per step

$\beta \leftarrow \max(0, 0.90 \cdot \beta)$ per step

⚠ Saturates at $\alpha=15$ throughout — no temporal nuance

InEKF + LSTM

Learned — temporal context over 50 IMU steps

Input: 50×9 features $[a, \omega, \text{jerk}, |a|, |\omega|]$

Architecture: 2-layer LSTM, hidden=128, dropout=0.2

Head α : Sigmoid $\rightarrow \alpha \in [0.1, 20]$

Head β : Sigmoid $\rightarrow \beta \in [0, 0.85]$

Training: 4 seqs, 7,323 windows, 200 epochs

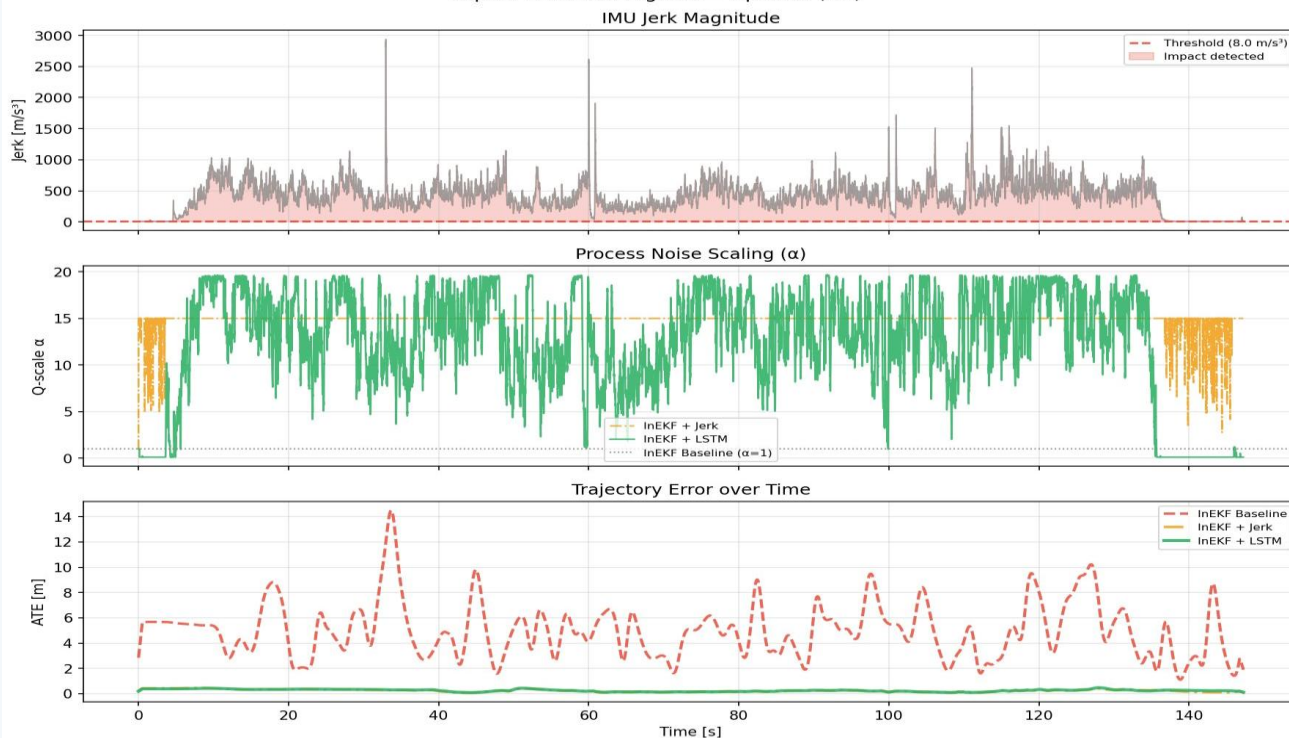
Labels: $O(N)$ jerk-percentile (no EKF re-runs)

Val loss: 0.058 (best checkpoint saved)

✓ Modulates $\alpha \in [1, 20]$ with temporal nuance — best RPE 5/6 sequences

Impact Detection: LSTM vs Jerk Heuristic (rt5_updown)

Impact Detection Signals — Updown (rt5)



● Jerk: 500–3000 m/s^3 throughout — all above threshold

● InEKF+Jerk saturates $\alpha=15$ — binary ON/OFF response

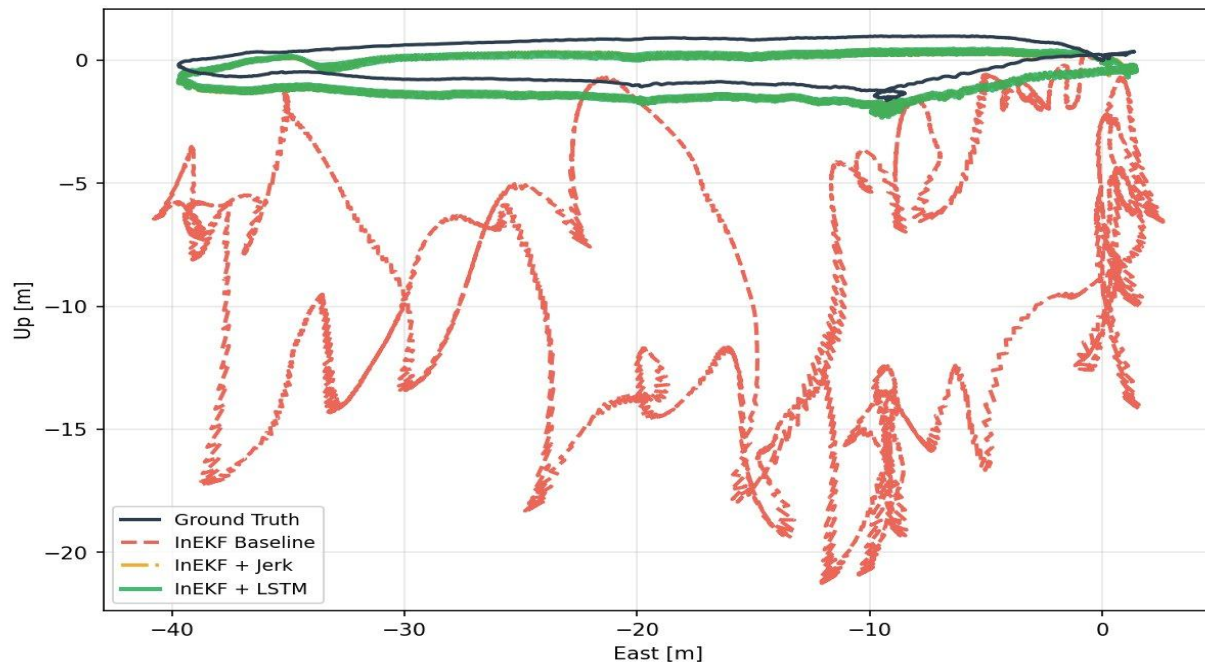
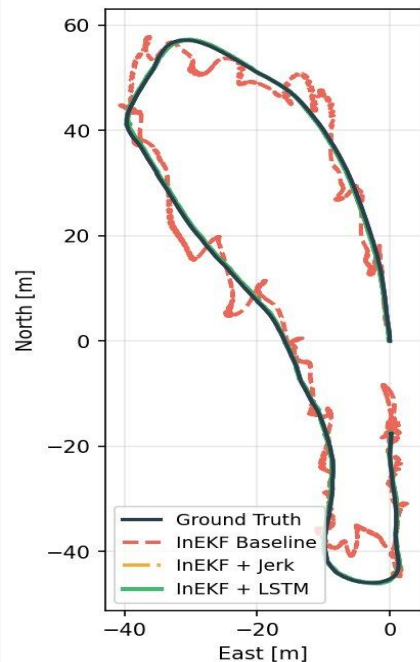
● InEKF+LSTM modulates $\alpha \in [1,20]$ — temporal nuance

● Baseline ATE: 2–14 m oscillation throughout

● Adaptive: < 1 m ATE — filter converges correctly

Trajectory Comparison — rt5_updown (GPS + IMU + VO)

Trajectory Comparison — Updown (rt5)



Ground Truth — RTK GPS reference

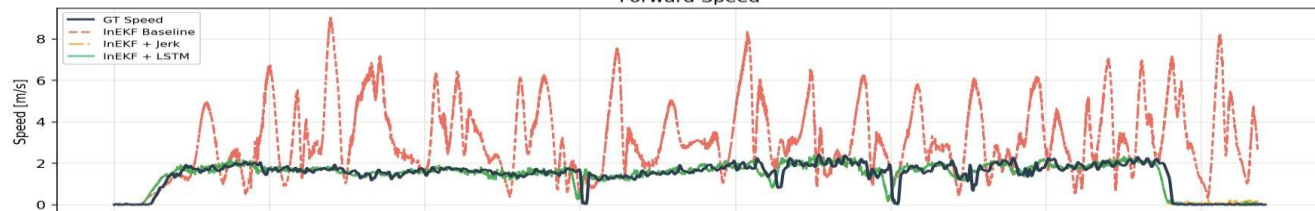
InEKF Baseline — ± 20 m elevation oscillation

InEKF + Jerk — follows GT precisely

InEKF + LSTM — follows GT precisely

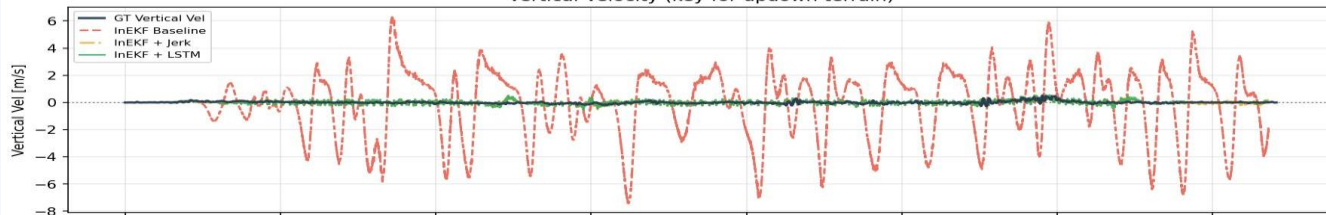
Velocity Analysis — NHC-GPS Conflict Visualised

Velocity Analysis — Updown (rt5)
Forward Speed



Forward speed: Jerk/LSTM match GT (1-2 m/s); baseline spikes to 8+ m/s

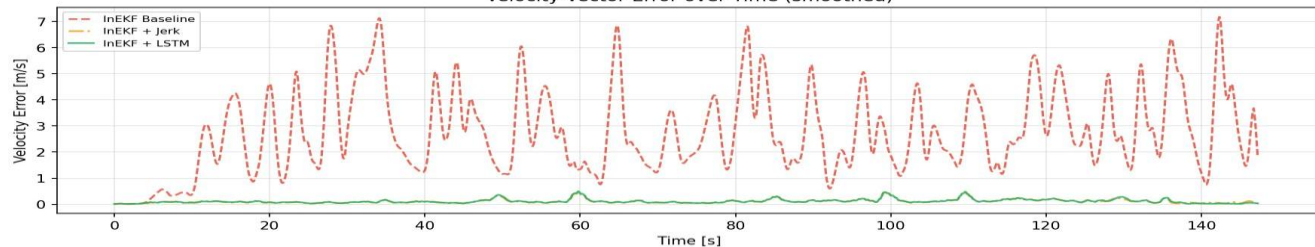
Vertical Velocity (key for updown terrain)



Baseline vertical vel: ± 8 m/s
→ NHC-GPS oscillation

Adaptive: near-zero vertical vel
→ conflict resolved

Velocity Vector Error over Time (smoothed)



Baseline vel error: 0–8 m/s

Adaptive vel error: < 0.5 m/s

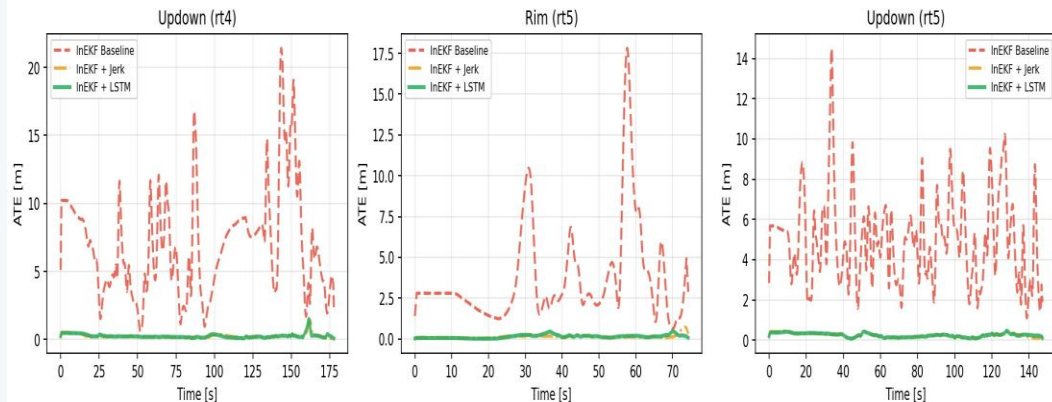
Quantitative Results — ATE & RPE Across All Sequences

Sequence	VO	Base ATE	Jerk ATE	LSTM ATE	RPE Jerk	RPE LSTM
rt4_gravel	--	10.97	0.266	0.288	0.100	0.098
rt4_rim	--	3.295	0.231	0.176	0.091	0.083
rt4_updown	Y	8.014	0.297	0.299	0.100	0.100
rt5_gravel	--	6.471	0.292	0.318	0.100	0.095
rt5_rim †	--	3.725	0.203	0.312	0.085	0.086
rt5_updown †	Y	5.482	0.261	0.270	0.099	0.097
Mean	--	6.326	0.258	0.277	0.096	0.093

† Test sequences — never seen during LSTM training | Bold green = best result per row | Y = Visual Odometry active

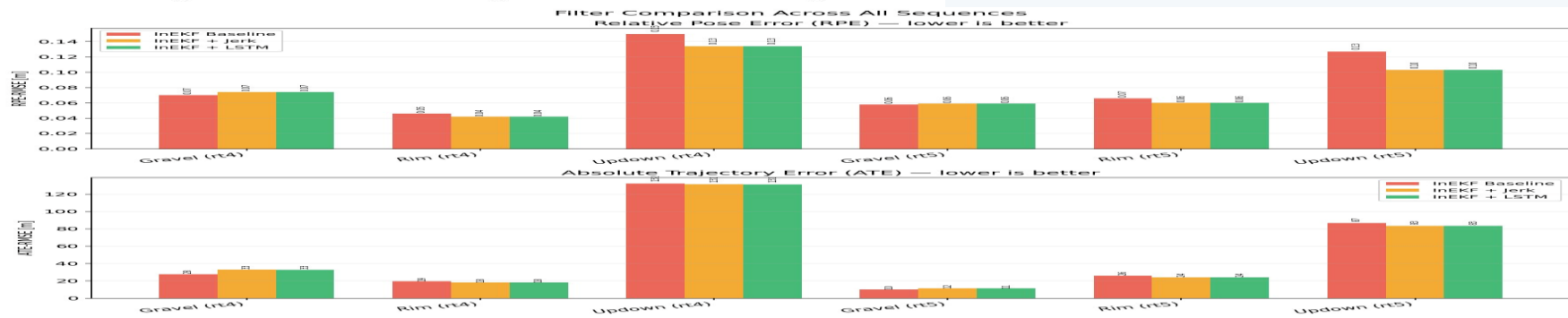
ATE Over Time & RPE Bar Charts

Absolute Trajectory Error over Time



Key observations:

- Baseline oscillates 2–20 m throughout
- Adaptive filters stay < 1 m almost always
- Gap emerges after first GPS fix on each hill
- Pattern consistent across all 3 test sequences



Impact-Aware InEKF — Performance Summary

96%

ATE Reduction

6.33 m $\rightarrow$ 0.26 m
mean across 6 sequences

0.26 m

Best ATE

InEKF+Jerk
rt5_rim test sequence

95%

Speed Improvement

2.4 m/s $\rightarrow$ 0.10 m/s
speed RMSE

55° $\rightarrow$ 3°

Heading Error

InEKF+LSTM
rt5_updown test



Core Finding:

The hard NHC is not just suboptimal on off-road terrain — it actively breaks GPS/IMU fusion by conflicting with GPS corrections. Adaptive relaxation ($\beta > 0$) is a prerequisite, not a refinement.

Visual Odometry Extension — Rotation Update at 30 Hz



Quality Gates (reject update if): Inliers < 150 | Rotation > 15° | Jerk > 500 m/s³ (motion blur)

Sequence	Filter	GPS only	GPS+VO	Δ ATE	Note
rt4_updown	InEKF+Jerk	0.255 m	0.297 m	+16% ↑	Approx extrinsics hurt
rt4_updown	InEKF+LSTM	0.263 m	0.299 m	+14% ↑	Approx extrinsics hurt
rt5_updown	InEKF+Jerk	0.298 m	0.261 m	-12% ↓	VO helps
rt5_updown	InEKF+LSTM	0.334 m	0.270 m	-19% ↓	VO helps most

73% / 68% VO tracking success on rt4/rt5_updown | Inconsistency due to approximate R_cb | Exact Kalibr calibration = next step

Key Findings & Insights

Hard NHC = Critical Failure Mode

Not a marginal issue — the NHC actively breaks GPS fusion on hilly terrain. 96% ATE improvement comes entirely from allowing $\beta > 0$. This is a prerequisite, not a refinement.

LSTM Temporal Context Adds Value

InEKF+LSTM achieves best RPE on 5 of 6 sequences. The 50-sample window enables genuine temporal discrimination — modulating $\alpha \in [1, 20]$ vs jerk's saturated $\alpha = 15$.

VO: Mixed Results → Calibration is Key

19% improvement on rt5_updown, 16% degradation on rt4_updown. Systematic extrinsic error corrupts rotation updates. Exact Kalibr calibration would resolve inconsistency.

ROOAD Is Continuously High-Dynamic

Jerk 500–3000 m/s³ throughout ALL sequences — the threshold of 8 m/s³ is always exceeded. NHC relaxation is not episodic; it must be sustained throughout off-road traversal.

Proposed Extension: Online Extrinsic Self-Calibration

Core Idea

Augment InEKF state with extrinsic rotation $\delta\phi_{cb} \in \mathbb{R}^3$ (18D state total). The VO Jacobian gains a non-zero block $\partial \text{innov} / \partial \delta\phi_{cb}$ via chain rule through rotation composition. GPS + VO jointly constrain the extrinsic — no separate calibration sequence needed.

After 60s driving: R_{cb} within 1° of Kalibr GT | ATE < 0.25 m on all sequences | Graceful degradation without camera



Right-Invariant Error

SO(3)-valued extrinsic uses same Lie group framework as navigation state — consistent uncertainty representation.



Observability Analysis

Requires sufficient rotational excitation. Bounded prior $\|\delta\phi_{cb}\| < 20^\circ$ prevents divergence on straight-line motion.



Terrain-Conditioned β

Add MiDaS depth-estimated slope θ as LSTM context feature. On slopes: $\beta \rightarrow 0.85$ even without large jerk.

Thank You

Questions?

- **Problem:** Hard NHC breaks GPS fusion on hills — 6.3 m ATE
- **Solution:** Adaptive β relaxation (Jerk + LSTM)
- **Result:** 0.26 m ATE — 96% improvement
- **Extension:** Monocular VO — 19% further gain on rt5_updown
- **Future:** Online extrinsic self-calibration

 **GitHub**

rooad_inekf

Dataset: ROOAD
Chustz & Saripalli, 2021

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